Discovery and Clinical Decision Support for Personalized Healthcare

Jinsung Yoon, C. Davtyan, M. van der Schaar

Journal:	Publication Year:	Citations:	DOI:
IEEE Journal of Biomedical and Health Informatics	2017.0	79	10.1109/jbhi.2016.2574857

CLINICAL & AGENTIC AI TAKEAWAY

The discovery engine (DE) effectively discovers patient characteristics relevant for diagnosis and treatment recommendations, improving healthcare outcomes.

ABSTRACT

With the advent of electronic health records, more data are continuously collected for individual patients, and more data are available for review from past patients. Despite this, it has not yet been possible to successfully use this data to systematically build clinical decision support systems that can produce personalized clinical recommendations to assist clinicians in providing individualized healthcare. In this paper, we present a novel approach, discovery engine (DE), that discovers which patient characteristics are most relevant for predicting the correct diagnosis and/or recommending the best treatment regimen for each patient. We demonstrate the performance of DE in two clinical settings: diagnosis of breast cancer as well as a personalized recommendation for a specific chemotherapy regimen for breast cancer patients. For each distinct clinical recommendation, different patient features are relevant; DE can discover these different relevant features and use them to recommend personalized clinical decisions. The DE approach achieves a 16.6% improvement over existing state-of-the-art recommendation algorithms regarding kappa coefficients for recommending the personalized chemotherapy regimens. For diagnostic predictions, the DE approach achieves a 2.18% and 4.20% improvement over existing state-of-the-art prediction algorithms regarding prediction error rate and false positive rate, respectively. We also demonstrate that the performance of our approach is robust against missing information and that the relevant features discovered by DE are confirmed by clinical references.

Study Overview & Background

With the advent of electronic health records, more data are continuously collected for individual patients, and more data are available for review from past patients. Despite this, it has not yet been possible to successfully use this data to systematically build clinical decision support systems that can produce personalized clinical recommendations to assist clinicians in providing individualized healthcare. In this paper, we present a novel approach, discovery engine (DE), that discovers which patient characteristics are most relevant for predicting the correct diagnosis and/or recommending the best treatment regimen for each patient. We demonstrate the performance of DE in two clinical settings: diagnosis of breast cancer as well as a personalized recommendation for a specific chemotherapy regimen for breast cancer patients. For each distinct clinical recommendation, different patient features are relevant; DE can discover these different relevant features and use them to recommend personalized clinical decisions. The DE approach achieves a 16.6% improvement over existing state-of-the-art recommendation algorithms regarding kappa coefficients for recommending the personalized

chemotherapy regimens. For diagnostic predictions, the DE approach achieves a 2.18% and 4.20% improvement over existing state-of-the-art prediction algorithms regarding prediction error rate and false positive rate, respectively. We also demonstrate that the performance of our approach is robust against missing information and that the relevant features discovered by DE are confirmed by clinical references.

Methodology & Clinical Focus

Study Classification: theoretical, modeling, or simulation study. Evaluated under orthopedic machine learning and artificial intelligence decision frameworks.

Key Findings & Analysis

The discovery engine (DE) effectively discovers patient characteristics relevant for diagnosis and treatment recommendations, improving healthcare outcomes.

Clinical Decision Support & Synthesis

Artificial intelligence and machine learning applications in orthopedic surgery demonstrate significant potential for standardizing clinical pathways, augmenting surgical decision-making, and personalizing post-operative rehabilitation. Continuous validation against prospective clinical datasets and multidisciplinary oversight remain paramount for ethical and safe clinical deployment.

Bibliographic Details & Indexing

- **Authors:** Jinsung Yoon, C. Davtyan, M. van der Schaar
- **Journal / Venue:** IEEE Journal of Biomedical and Health Informatics (2017.0)
- **SJR Metric:** Quartile 1.0
- **Citation Count:** 79 citations recorded
- **Digital Object Identifier (DOI):** https://doi.org/10.1109/jbhi.2016.2574857
- **Consensus Evidence Link:** https://consensus.app/papers/discovery-and-clinical-decision-support-for-personalized-yoon-davtyan/6a9fb394cffd5fe48a4ddacda761ec6d/

Source & Provenance: Curated for Agentic Orthopedics Research Collection • DOI: https://doi.org/10.1109/jbhi.2016.2574857 • Consensus Page: https://consensus.app/papers/discovery-and-clinical-decision-support-for-personalized-yoon-davtyan/6a9fb394cffd5fe48a4ddacda761ec6d/ • Verified and rendered via Deep Search & Academic Repositories.